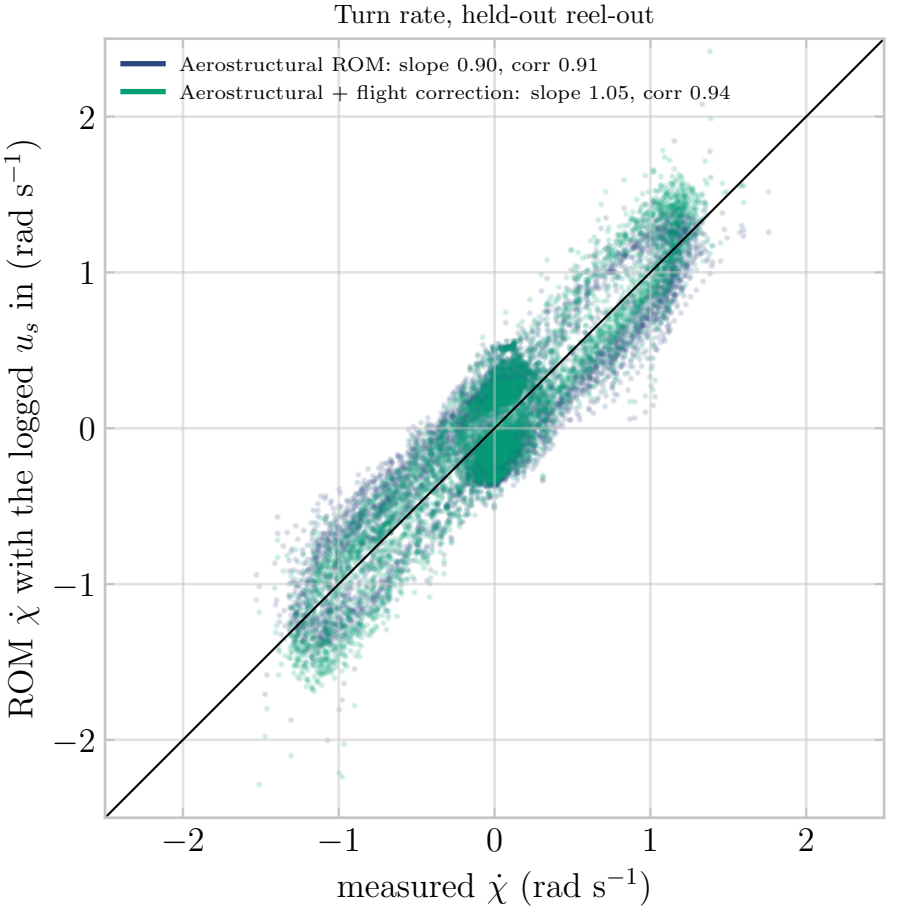
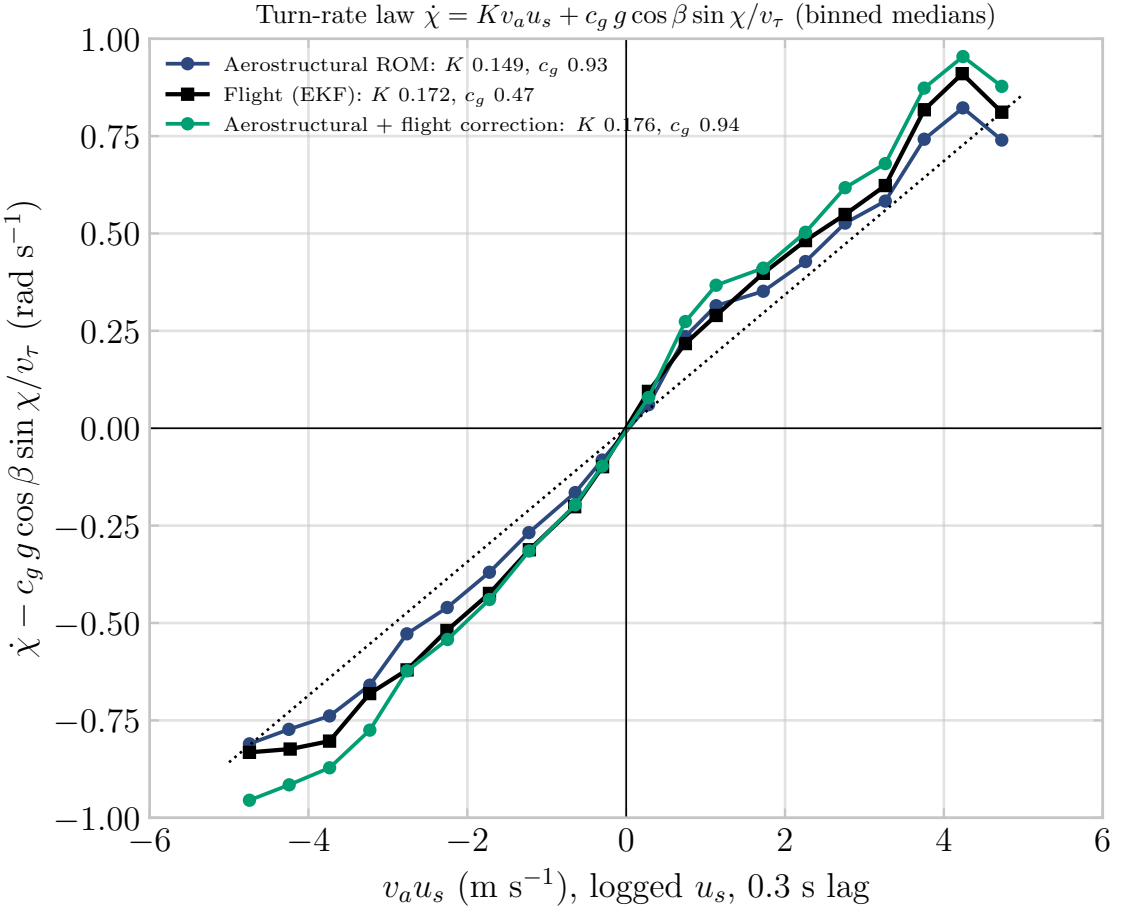

Turn rate, held-out reel-out
Aerostructural ROM: slope 0.90, corr 0.91
Aerostructural + flight correction: slope 1.05, corr 0.94
ROM $\dot{\chi}$ with the logged u_s in (rad s^{-1})
measured $\dot{\chi}$ (rad s^{-1})
2
1
0
−1
−2
−2
−1
0
1
2


Turn-rate law $\dot{\chi} = K v_a u_s + c_g\, g \cos\beta \sin\chi / v_\tau$ (binned medians)
Aerostructural ROM: K 0.149, c_g 0.93
Flight (EKF): K 0.172, c_g 0.47
Aerostructural + flight correction: K 0.176, c_g 0.94
$\dot{\chi} - c_g\, g \cos\beta \sin\chi / v_\tau$ (rad s^{-1})
$v_a u_s$ (m s^{-1}), logged u_s, 0.3 s lag
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−6
−4
−2
0
2
4
6